



ENGINEERING DOCUMENTATION

DOCUMENT 08 / 09

Release Summaries / Changelog

AI Hiring Platform · Intelligence-First

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Lexy — Release Summaries / Changelog

Engineering documentation set · Doc 8 of 9 What has been built, grouped by theme, newest-first. This is a curated engineering changelog (the “what shipped and why it mattered”), reconstructed from git history. It is not a per-commit dump — see `git log` for that.

■ Intelligence Engine — the “9-gap” credibility arc (most recent major work)

This arc systematically closed the gaps that stood between “AI scores” and *defensible, self-improving* AI scores. In rough order:

Theme	What shipped	Why it mattered
Fairness & adverse impact	4/5ths analysis on (candidate, job) units, demographics firewall, <code>FAIRNESS_DIRECTIVE</code> + PII redaction on every scorer, <code>test:fairness</code> suite.	Legal/ethical floor for hiring AI; makes the product sellable into regulated hiring.
Outcome labels & quality-of-hire	Capture of hire/reject outcomes + post-hire pulses (30/90d).	Creates the <i>training signal</i> the learning loop needs.
Model versioning & backtest harness	<code>backtest.ts</code> re-runs history through a candidate config; only promotes if F1/precision improves.	Scoring changes must <i>earn</i> production; no silent quality regressions.
Thin-data honesty & calibration	Confidence breakdown, hard caps when critical signals missing, calibration labeling.	Honest uncertainty = trust = the whole “defensible” pitch.
Learned scoring	<code>learned-scoring.ts</code> per-tenant weights, sample-gated (≥ 25) + backtest-gated, never-throw fallback.	Each tenant’s model adapts to its own outcomes.
Similar-hire embedding signal	Real embedding-based “looks like your past good hires” signal, backtested vs the LLM-vs-ICP baseline.	Stronger Fit signal grounded in the tenant’s actual hires.
Cross-tenant global prior	Network-effect prior from pooled <i>sufficient statistics</i> (never raw rows); thin tenants shrink toward it.	The moat: product gets smarter the more it’s used, privacy intact.
Sourcing provider adapter layer	Pluggable provider registry (PDL/SERP/GitHub/internal); env-disable; never-throw.	External sourcing becomes safe to extend and operate.

Follow-up fix (latest): provider runs now race a **per-kind timeout** (discovery 20s / enrichment 90s, env-overridable) so a *hung* provider can't block the whole sourcing fan-out — graceful degradation previously only caught throws, not hangs.

■ Interview pipeline — reliability at scale

A long series of fixes hardening the highest-stakes candidate moment:

- **Azure speech scaling:** dedicated TTS vs STT accounts, account **pooling + admission control**, per-account/per-format circuit breaker — fixes concurrent-load STT latency.
- **TTS watchdog:** `speakText` always settles before the mic opens — fixes candidates frozen on Q1 (“Lexy not responding”) on mobile.
- **Mobile STT path:** mic-only segments routed to server `/transcribe` (browser Web Speech can't run while the video recorder holds the mic); typing fallback when the mic isn't heard.
- **STT quality telemetry:** persisted `stt_transcribe_events` (survive restarts), automatic alerting on mobile transcription regressions, per-provider/per-format breakdown, admin **Transcription Health** dashboard, automatic retention/cleanup.
- **Recording reliability:** resumable S3 multipart uploads, keepalive save on unload, end-of-interview detection, per-question scoring at `/end`.
- **Audio-format routing** covered by automated tests; verified on real iOS/Android.

■ Tenant isolation & data integrity

- RLS tenant-isolation hardening; documented data-leak surfaces and the subtree-visibility model.
- Candidate-score **source-of-truth** consistency (accrued intelligence wins; KPI counts match the lists they link to).
- Candidate **dedup** (one row per tenant+lower(email)); orphan intelligence-row resolution (no more “Unknown” cards).
- DNC exclusion across kanban / sourcing / candidate list with live cache invalidation.

■ Outreach, inbox & engagement

- Multi-step drip outreach (campaigns, enrollments, sequences, autopilot) with AI-personalized drafts; **approval-required by default**, edit-before-approve.
- GenAI email auto-reply + 24h interview-invite flow with secure invite tokens.
- One-click candidate self-decline → auto-reject + DNC.
- Recruiter inbox with inbound-email inline-image rendering (SendGrid Inbound Parse + SES→SNS), reply classification, re-engagement agent.
- Anti-ghosting detection + nurture; weekly/recruiter digests; LinkedIn monitor.

■ Connection Engine (additive, feature-flagged)

Status: dormant by default — treat as experimental scaffolding, not live product, unless a flag is explicitly on. Both flags default to **off** (the code activates only when the env var equals the string `"true"`), and they are **not enabled in the current environment**. So the routes/logic are merged and ready but inert: no tenant gets Connection Engine behaviour until someone sets the flag. If you're touching this area, assume it's *off in production* and verify the flag before assuming any of it is running for real users.

- Employer-side `connectionStrengthScore` (0–100) from engagement events, fully isolated from core scoring (`ENABLE_CONNECTION_ENGINE`).
- Candidate-side co-pilot: hiring momentum, connection strength, next-best-action (`ENABLE_CANDIDATE_CONNECTION_ENGINE`).

■ Monetization & multi-tenant business model

- Country-level **subscription module**: display pricing + automatic lifecycle (trial expiry, grace, reminders) via `paid_through` + grace windows.
- Stripe checkout for upgrades; self-serve email-verified trial signup.
- India pricing model + unit-economics modeling artifacts.
- 3-level tenant hierarchy (platform admin → enterprise/agency → branch/sub-client) with branch/sub-client management.

■ AI infrastructure

- **AI job queue** (`ai_jobs`): DB partial-unique-index dedupe, background worker with `FOR UPDATE SKIP LOCKED` , `enqueueAiJob` returns null on failure.
- 10-agent orchestrator with canonical `AGENT_ORDER` ; every run feeds the intelligence layer.
- AI governance: kill switch, tenant scoping, atomic-send, build-breaking `check:no-adverse-writes` guard.

■ How to read the raw history

```
git --no-optional-locks log --oneline          # full commit record
git --no-optional-locks log --oneline <file> # history for one file
```

The recent Intelligence-Engine arc corresponds to a sequence of scoped tasks; this document summarizes their *outcomes and rationale* so you don't have to reconstruct intent from commit messages.